



8/13/2026

ASSIGNMENT DAY 4

Component Level Hardware
Troubleshooting & Diagnostics.

TASK 1.

- ❖ **Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating then document each step you took and any issues you observed during the process.**

➤ **RAM Reseating Procedure:**

- **Locate the RAM:** I opened the PC and identified the RAM slots on the motherboard.
- **Remove the RAM:** I released the side clips and carefully removed the RAM module.
- **Re-insert the RAM:** I aligned the notch and pressed the module firmly until the clips locked.
- **Check the System:** I powered on the PC and checked whether it started normally.
- **Issue Observed:** No major issue was observed, and the RAM was detected normally.

TASK 2.

- ❖ **Simulate Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix. Think about display issues, driver checks, and physical connections.**

➤ **GPU Troubleshooting Checklist:**

- **Identify the Symptom:** Check for no display, screen artifacts, crashes, or poor graphics performance.
- **Check Physical Connections:** Make sure the GPU is properly seated and its power cables are firmly connected.
- **Check the Display:** Test the monitor, display cable, and correct input source.
- **Check GPU Drivers:** Update or reinstall the graphics driver if it is outdated or corrupted.
- **Confirm the Fix:** Restart the PC and run a game or graphics test to ensure the GPU works normally.

TASK 3.

❖ **Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.**

➤ **Storage Issue Simulation:**

- **Before disconnecting:** The BIOS normally detects the NVMe SSD and shows it under the storage information.
- **After disconnecting:** The BIOS no longer detects the SSD, and the system may show **“No Bootable Device”** or **“Boot Device Not Found.”**
- **After reconnecting:** The NVMe SSD appears again in the BIOS storage list.
- **Final Check:** The laptop boots normally once the drive is detected again.

➤ **Issue Observed:** The error was caused by the storage drive being disconnected or not properly detected.

TASK 4.

❖ **Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.**

➤ Troubleshooting a PC That Randomly Restarts:

🔧 **Component to Check First:** RAM

➤ **Reason:** Faulty or poorly seated RAM can cause sudden restarts, crashes, and system instability.

🔧 **Diagnostic Steps:**

➤ **Reseat the RAM:** Remove and firmly reinstall the RAM module.

➤ **Check One Stick at a Time:** Test each RAM stick separately to find a faulty module.

➤ **Inspect the RAM Slot:** Check for dust, damage, or loose connections.

➤ **Test the System:** Restart the PC several times and check for further random restarts.